Project V: Advanced Software Development Tools

Design Sprint Input

Purpose

The purpose of this document is to plan out your goals for the given design sprint. So this document will be used a total of 3 times.

Overview

Each design sprint will plan to (a) fix/enhance the previous code and (b) add new functionality into the system. The given requirements/issues are already defined and released, so it's a matter of selecting the requirements you plan to complete this week, assigning them to a "Sprint Iteration" in the collaborative SRS, and reviewing any issues raised by the class on your service functionality.

What to do

Step #1: Fixes/Enhancements

In the text box below provide details about any fixes or enhancements you plan to complete during this design sprint. Make sure you reference Github Issue numbers for traceability. If they are enhancements that require a change in requirements, these need to be reviewed by the class before proceeding.

```
MODULE 10 - ATTENDANCE & PARTICIPATION TRACKING
Sprint 1 | Prepared October 16, 2026
Due October 20, 2026, 11:59 PM America/Toronto
Repository: https://github.com/Arda190777/C3L
Branch: feature/attendance-sprint-1
Milestone: Module 10 - Sprint 1 - milestone/1

PLANNED FIXES
No prior attendance implementation or reported attendance defect exists in the

PLANNED ENHANCEMENTS
#3: Replace demo-only behavior with server validation, escaped HTML, safe SQ
errors. Reject blank/oversized IDs, impossible dates, invalid statuses and long
#4: Add unit/integration/regression evidence; prepare design/API documents an
Module 10 and add no dependencies.

TASK OWNERSHIP
Arda Efe Cetintav: #4, small form UI and final PDF/release checks.
Jaspreet Gakhal: #9, small history UI, demo and use-instruction review.
Gera Paoli: #1/#3, saving, validation, SQLite and unit/storage tests.
Nasir Ishaq Sani: #2/#8, queries, filters, API, regression and SRS.
Gera/Nasir's GitHub account assignments are deferred for now. Team verificati

COMMON SRS / DESIGN
The supplied brief has no numbered requirements. Proposed IDs and Sprint Iter
class SRS link was not supplied; the shared SRS update and official ID mappin
unchanged.

Issues: https://github.com/Arda190777/C3L/issues
```

Step #2: New Functionality

In the text box below, provide details about the new functionality you plan to design and implement during this design sprint. Make sure you reference the requirement numbers you will be focusing on for traceability.

SPRINT 1 GOAL
Record and correct daily attendance, distinguish absence types and retriev

SELECTED NEW FUNCTIONALITY
M10-FR-01 - Record attendance - #1: enter course ID, student ID, date, st
course/student/date. Re-saving corrects that mark. Persist records in loca

M10-FR-02 - Attendance history - #2: list newest date first; filter by exact
dates; show a useful empty state. Provide both HTML and JSON history.

M10-FR-03 - Excused/unexcused absences - #1, #3: support Present, Abse
Reject other statuses.

TRACEABILITY NOTE
These are proposed local IDs, not verified class SRS numbers. They map t
of the supplied project brief. Copy/map them into the collaborative SRS wi
#8.

ACCEPTANCE / TESTING
Validate IDs - 1-50 characters, real dates and notes - up to 300 characters
Escape displayed input. Verify file persistence, form save/correction, filte
integration. Record real regression results.

IMPLEMENTATION
JavaScript + Elysia + Bun native SQLite; entry index.js at /10-attendance.
candidate passes 25 module tests and 81 full-repo tests.

DEFERRED
#5: identity roles and enrolled course/student validation. Sample IDs only
#6: M10-FR-04 participation, M10-FR-05 reports, M10-FR-06 statistics.

SUBMISSION / TEAM WORK
Submit completed Input and Release Note PDFs plus the Doxygen HTML ZI
the demo remain pending.

Rubric

See eConestoga.

What to Hand In

Please submit the following files to eConestoga for marking

- A completed copy of this PDF document
- A ZIP containing the Doxygen generate HTML Design Information.

Make sure you have assigned all your Issues planned to fix in this issue in Github, have updated the collaborative SRS to assign the requirements to the sprint and created tasks in your Github for new enhanicements.